Reducing the incidence of twins and triplets.

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Multiple pregnancy rates remain high after assisted conception because of a misconceived assumption that transferring three or more embryos will maximize pregnancy rates. Maternal morbidity is sevenfold greater in multiple pregnancies than in singletons, perinatal mortality rates are fourfold higher for twins and sixfold higher for triplets, while cerebral palsy rates are 1-1.5% in twin and 7-8% in triplet pregnancies. Therefore, multiple pregnancies must be considered a serious adverse outcome of assisted reproductive techniques. Primary prevention of multiple pregnancies is the solution. The overwhelming evidence presented in this chapter demonstrates that limiting the embryo transfer in in vitro fertilization to two embryos would significantly reduce adverse maternal and perinatal outcomes by reducing the incidence of high order multiple pregnancies without reducing take-home-baby rates. Secondary prevention by multifetal pregnancy reduction is effective, but not acceptable to all patients. New developments in blastocyst culture, single embryo transfer, embryo cryopreservation and pre-implantation aneuploidy exclusion, should allow improvements in pregnancy rates without increasing multiple pregnancies.